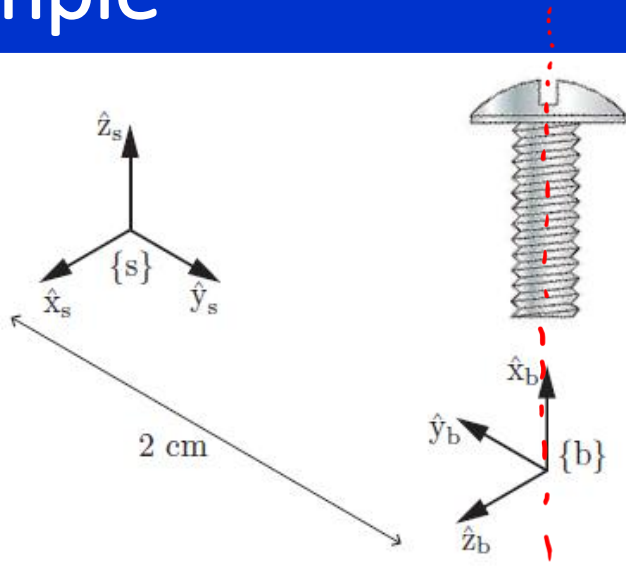


Example



- screw is driven into a tapped hole by a screwdriver at @ 3 rad/sec
- $\hat{x}_b$ - axis is along the axis of the screw
- $\hat{z}_s$ is aligned w/ $\hat{x}_b$
- origin of $\{s\}$ is displaced 2 cm along the $\hat{y}_b$ - axis

Figure 3.2: As the machine screw goes into a tapped hole, it advances linearly by 4π mm every full rotation of the screw.

What are V_b , S_b , V_s , and S_s ?

For right-handed threads looking down (rotates clockwise when advancing into a tapped hole)

$\uparrow \hat{x}_b$
 $\downarrow \hat{s}$

$$\omega_b = [-3 \text{ rad/sec}, 0, 0]^T$$

pitch $h = 4\pi \text{ mm} / 2\pi = 2 \text{ mm/rad}$

As the screw is rotated at 3 rad/sec, it moves linearly in the $-\hat{x}_b$ direction

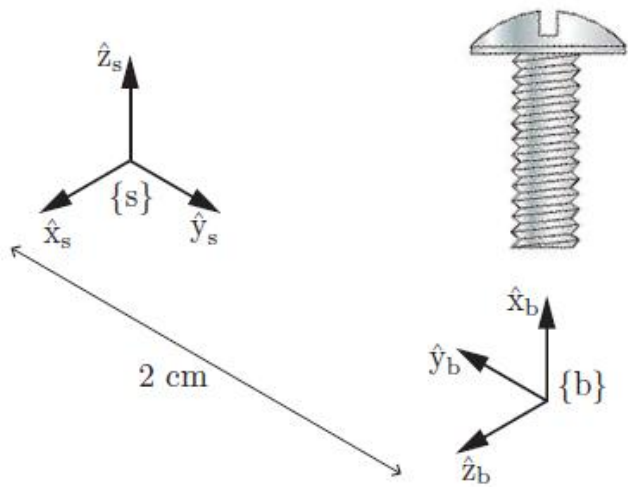


Figure 3.2: As the machine screw goes into a tapped hole, it advances linearly by 4π mm every full rotation of the screw.

$$v_b = (2 \text{ mm/rev})(3 \text{ rev/sec}) = -6 \text{ mm/sec} \hat{x}_b$$

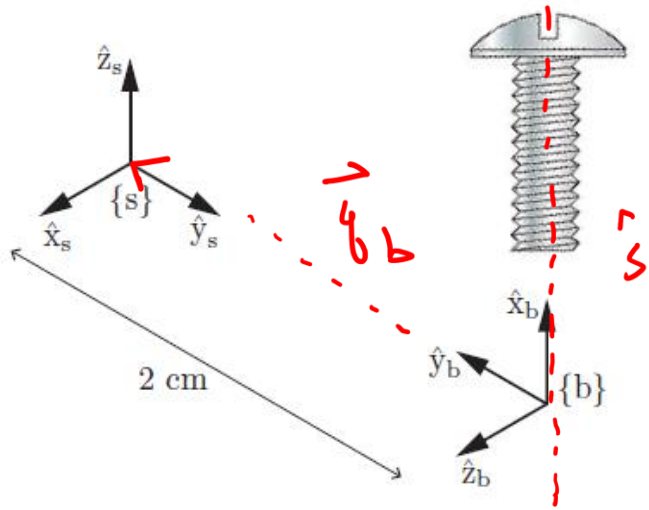
$$= [-0.006 \text{ m/sec}, 0, 0]^T \text{ in SI units}$$

$$V_b = [\omega_b, v_b]^T = [-3, 0, 0, -0.006, 0, 0]^T$$

$$s_b = v_b / \|\omega_b\| \text{ where } \|\omega_b\| = \sqrt{(-3)^2 + 0 + 0} = 3$$

$$= [-1, 0, 0, -0.002, 0, 0]^T$$

$$V_b = s_b \dot{\theta} \text{ where } \dot{\theta} = \|\omega_b\| = 3$$



In the $\{s\}$ frame, the axis of rotation is aligned w/ the $-\hat{z}_s$ -axis

$$\omega_s = [0, 0, -3]^T$$

Figure 3.2: As the machine screw goes into a tapped hole, it advances linearly by 4π mm every full rotation of the screw.

For a point at the origin of $\{s\}$, rigidly attached to the screw

- ① downward motion of the screw: $(0, 0, -0.006)$
- ② linear component in the $-\hat{x}_s$ direction from the rotation of the screw (or $-\omega_s \times q_s$)

In $\{b\}$ coordinates, the point at the origin of $\{s\}$ is

$$q_b = (0, 0.02, 0)$$

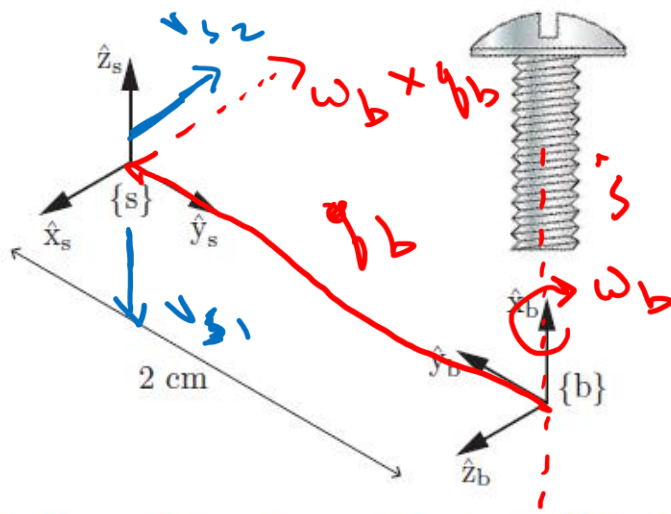


Figure 3.2: As the machine screw goes into a tapped hole, it advances linearly by 4π mm every full rotation of the screw.

So, linear motion of a point at $\{s\}$ due to rotation of the screw is

$$\omega_b \times q_b = (0, 0, -0.06)$$

In the $\{s\}$ frame, this is $(-0.06, 0, 0)$

So, the total linear motion at $\{s\}$ expressed in $\{s\}$ is

$$\begin{aligned} v_s &= \underbrace{(0, 0, -0.006)}_{s_1} + \underbrace{(-0.06, 0, 0)}_{s_2} \\ &= (-0.06, 0, -0.006) \end{aligned}$$

$$V_s = (0, 0, -3, -0.06, 0, -0.006)$$

Example

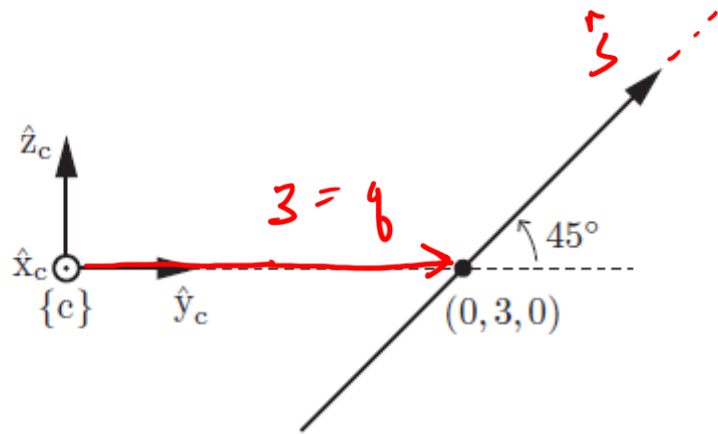


Figure 3.3: A screw axis in the $(\hat{y}_c, \hat{z}_c)$ plane.

If the pitch of the screw is $h=10$ linear units per radian, what is the screw axis S_c ?

$S_c = (S_{c_w}, S_{c_v})$ where S_{c_w} is a unit vector aligned with the axis of the screw

$$S_{c_w} = \hat{S} = (0, \cos 45^\circ, \sin 45^\circ) = (0, 1/\sqrt{2}, 1/\sqrt{2})$$

The linear components are $S_{c_v} = h\hat{S} - \hat{S} \times q$
 ↓ linear motion along axis of screw
 → linear component due to rotation about axis of screw

$$h\hat{S} = (0, \frac{10}{\sqrt{2}}, \frac{10}{\sqrt{2}})$$

$$-\hat{S} \times q = (\frac{3}{\sqrt{2}}, 0, 0)$$

$$S_{c_v} = (\frac{3}{\sqrt{2}}, \frac{10}{\sqrt{2}}, \frac{10}{\sqrt{2}})$$

Exponential coordinates of a homogeneous transformation

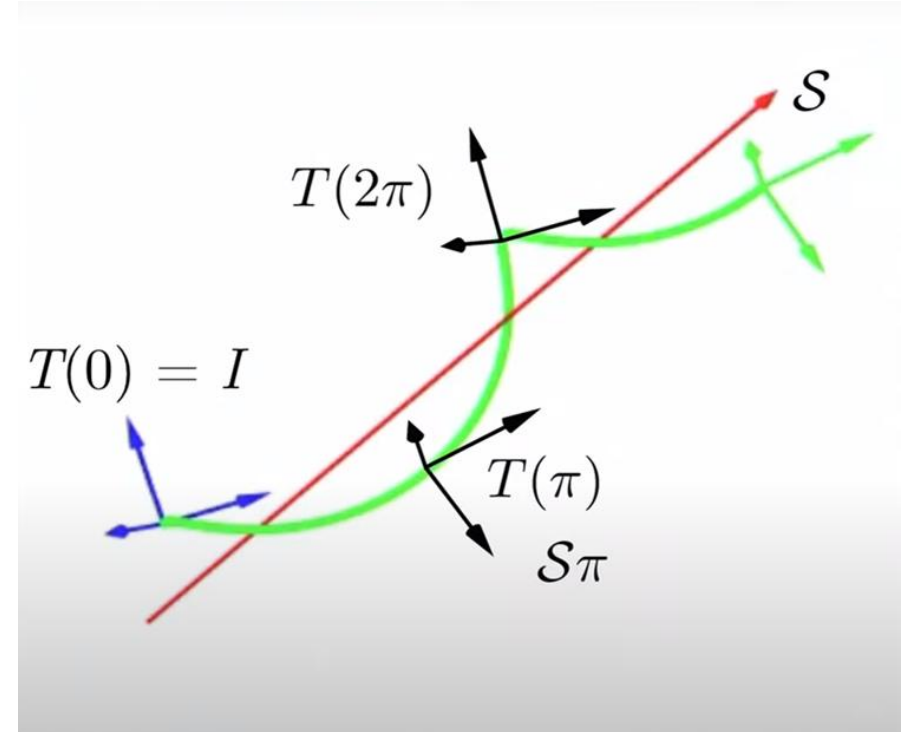
Chasles-Mozzi theorem: every rigid-body displacement can be expressed as a displacement along a fixed screw axis S in space.

Recall: the rotation matrix $R \in SO(3)$ can be represented in exponential coordinates $\theta \hat{\omega}$

The differential equation $\dot{p}(t) = [\hat{\omega}]p(t)$ has the solution

$$p(\theta) = \text{Rot}(\hat{\omega}, \theta)p(0)$$

$$p(\theta) = e^{[\hat{\omega}]\theta}p(0) \text{ from } t=0 \text{ to } t=\theta$$



By analogy, we define the 6-dimensional exponential coordinates of a homogeneous transformation $T \in SE(3)$ as $S\theta \in \mathbb{R}^6$:

S is the screw axis

θ is the distance that must be traveled along the screw axis to take a frame from the origin I to T

Parameterizing homogeneous transformations

We can create analogues for exponential coordinates from rotations to all rigid-body motions (transformations):

	Rotations	Rigid-Body Motions
Axis:	$\hat{\omega}$	$\mathcal{S} = (\omega, v)$
Exponential Coordinates:	$\hat{\omega}\theta \in \mathbb{R}^3$	$\mathcal{S}\theta \in \mathbb{R}^6$
Matrix Representation:	$[\hat{\omega}]\theta$	$[\mathcal{S}]\theta = \begin{bmatrix} [\omega] & v \\ 0 & 0 \end{bmatrix} \theta$ (4 x 4 matrix)

Exponential coordinates: 2 cases

Case 1: if $\omega = 0$ and $\|v\| = 1$, then the pitch is *infinite* and we have pure translation. θ is the linear distance traveled along the screw axis.

$$e^{[S]\theta} = \begin{bmatrix} I & v\theta \\ 0 & 1 \end{bmatrix}$$

Case 2: if $\|\omega\| = 1$, then θ is the angle of rotation about the screw axis.

$$e^{[S]\theta} = \begin{bmatrix} e^{[\omega]\theta} & (I\theta + (1 - \cos(\theta))[\omega] + (\theta - \sin(\theta))[\omega]^2)v \\ 0 & 1 \end{bmatrix}$$

Check out Section 3.3.3.1 of *Modern Robotics* for more details!

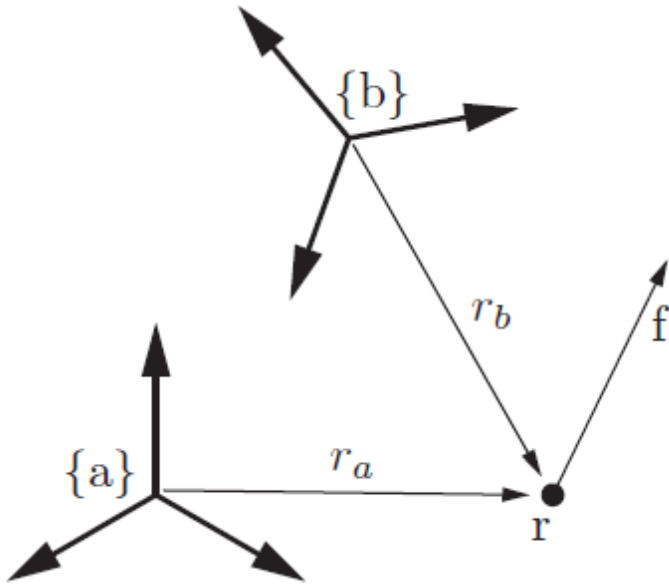
Wrenches

If a linear force $\vec{f}_a$ acts on a rigid body at point $\vec{r}_a \in \mathbb{R}^3$ in $\{a\}$, it exerts a moment $\vec{m}_a = \vec{r}_a \times \vec{f}_a$.

We can merge force and moment together: $F_a = \begin{bmatrix} \vec{m}_a \\ \vec{f}_a \end{bmatrix} \in \mathbb{R}^6$

This is known as a **wrench**. The total wrench on a body is the vector sum of all the individual wrenches. These wrenches must all be expressed in the same frame!

Representing wrenches in different frames



To represent wrenches in different frames, we use the fact that power is the same regardless of frame

$$V_b^T F_b = V_a^T F_a$$

$$V_b^T F_b = ([Ad_{T_{ab}}] V_b)^T F_a$$

$$V_b^T F_b = V_b^T [Ad_{T_{ab}}]^T F_a$$

$$\begin{aligned} F_b &= [Ad_{T_{ab}}]^T F_a \\ F_a &= [Ad_{T_{ba}}]^T F_b \end{aligned}$$

Summary

- Homogeneous transformation matrices $T \in SE(3)$ combine orientation and position and represent general rigid-body motion
- Any rigid-body motion can be achieved by starting from the fixed (home) reference frame and integrating a constant twist, or spatial velocity, for a specified time
- This motion resembles the motion of a screw -> rotating about and translating along the same fixed axis
- Every rigid-body displacement can be expressed as a displacement along a fixed screw axis $\mathcal{S}$; we call the 6-parameter representation of the configuration the **exponential coordinates**